Slow Conduction Corridors and Conduction Velocity Predict Atrial Fibrillation Recurrence After First Ablation

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Abstract

Atrial fibrillation (AF), the most prevalent sustained arrhythmia, is linked to an elevated risk of stroke, heart failure, and mortality. This study investigated whether conduction velocity (CV) measurements derived from pre-ablation electroanatomical maps (EAM) predict long-term AF recurrence following a first ablation procedure (n=45).

In univariate Cox analysis, a lower median CV was associated with a higher risk of recurrence (hazard ratio (HR): 2.08 per 0.1 m/s decrease, $p = 0.049$). Consistently, after dichotomization, a median CV < 0.66 m/s was significantly associated with an increased recurrence risk (HR: 4.69, $p = 0.016$). Slow conduction corridors (SCCs), defined as regions with CV < 0.5 m/s, provided additional prognostic value. Within SCCs, both a larger low-voltage zone area (HR: 5.11, $p = 0.010$) and greater unipolar signal fragmentation (HR: 5.20, $p = 0.009$) were significantly associated with AF recurrence.

Multivariate analysis confirmed median CV as an independent predictor of AF recurrence, alongside SCC-related structural and voltage features (HR: 8.25, $p = 0.007$, and HR: 6.37, $p = 0.011$, respectively).

These findings identify CV-based metrics as robust predictors of AF recurrence and support their clinical integration to enhance patient selection and ablation outcomes.

1. Introduction

Atrial fibrillation (AF) is the most common cardiac condition in adults, with its global prevalence projected to double in the coming decades due to population aging and advances in detection methods [1]. AF is associated with substantial morbidity and mortality, including heart failure and stroke [2]. Because the onset and progression of AF are often driven by comorbidities, optimal management relies on early and continuous control of these underlying factors [1].

Pulmonary vein isolation (PVI) currently represents the treatment of choice for AF. However, despite improved success rates in recent years, recurrences still occur, prompting the investigation of additional arrhythmogenic substrates such as the posterior wall or the left atrial appendage [2, 3]. Despite these strategies, AF recurs in 10% to 35% of patients after an initial ablation [4].

The mechanisms underlying atrial remodeling remain incompletely understood, making electroanatomical mapping (EAM) essential for characterizing the arrhythmogenic substrate. EAM-derived features have been proposed to guide ablation by identifying targets such as low-voltage areas, fibrotic regions, and rotational drivers [5], based on a detailed analysis of electrograms (EGMs). These recordings enable the quantification of relevant markers, including peak-to-peak voltage [6] and signal fractionation [7].

Atrial conduction velocity (CV) reflects both structural and functional myocardial properties [8] and represents a robust indicator of substrate integrity, with regions of slow conduction serving as potential ablation targets. Peak-to-peak voltage also provides complementary information on tissue viability, with reduced amplitudes typically indicating fibrotic or diseased myocardium.

However, neither CV nor voltage metrics alone fully capture the spatiotemporal behavior of the myocardium. To address this limitation, we propose integrating CV mapping with localized EGM analysis within slow conduction corridors (SCCs). By focusing on these functionally defined regions, which may reveal local electrophysiological abnormalities associated with AF recurrence, rather than on global atrial properties, this approach aims to achieve a more precise characterization of the arrhythmogenic substrate. The primary objective of this study was to evaluate the predictive value of these pre-ablation CV-derived markers, together with their local features, for long-term AF recurrence in patients undergoing a first PVI procedure.

2. Materials and Methods

2.1. Study Population

This study included EGM recordings acquired during EAM in 45 patients with paroxysmal or persistent AF (16 persistent) undergoing a *de novo* PVI procedure at Hospital Clínico Universitario Lozano Blesa (Zaragoza, Spain).

During a maximum 4-year follow-up, 9 patients (20%) experienced AF recurrence, defined per consensus guidelines (90-day blanking period). These included **blanking period** ($\leq$ 3 months, persistent beyond blanking; n=2), **early** (3-12 months; n=3), and **late** ($>$ 12 months; n=4) **recurrences**.

Three-dimensional atrial geometry and high-density mapping (HDM) data were acquired both before and after ablation using the Rhythmia mapping system and the INTELLAMAP ORION^{TM} basket catheter (Boston Scientific Corporation, Marlborough, MA). The present analysis focuses on pre-ablation maps, as they offer superior mesh quality and more complete atrial coverage compared to post-ablation maps. The recorded EGMs were sampled at 953.674 Hz, with a mean activation period of 177.1 ± 28.9 ms.

2.2. EGM Processing

Pulmonary veins were excluded from the atrial meshes according to the provided ablation lines, yielding maps with an average of 6,872 ± 1,491 mesh vertices and 3,586 ± 1,519 recorded EGMs. Signal processing was performed in MATLAB. EGMs were bandpass filtered between 30 and 300 Hz. For each patient, signals were temporally aligned using cross-correlation, and 20 ms were trimmed from both the beginning and end of the activation window to reduce noise and minimize potential errors in the marker computations described in Section 2.4.

2.3. Conduction Velocity Maps

CV was computed from the activation time map using the OpenEP toolbox [9]. This platform uses Radial Basis Function (RBF) interpolation to reconstruct the spatial activation sequence and derive the velocity vector field through analytical differentiation of the interpolated surface [10].

From these maps, several global metrics were extracted, including the median, 10th percentile (P_{10}), and 25th percentile (P_{25}) of all CV values. In addition, SCCs were identified using established clinical thresholds ($CV < 0.5$ m/s) [11].

SCC burden was quantified as the percentage of the total atrial surface area. To further characterize the underlying substrate, SCC regions were intersected with bipolar voltage maps to quantify the proportion of scar tissue (< 0.1 mV), border zone (0.1–0.5 mV), and the overall low-voltage zone (LVZ) (< 0.5 mV) within these regions.

2.4. SCC Markers

To minimize interpolation bias and enhance substrate characterization, SCC tissue properties were additionally assessed using local EGMs. For each patient, all signals within 2 mm of the SCC boundaries were selected, yielding an average of 1,748 ± 1,575 EGMs per patient. From these signals, the following markers were computed:

(1) **Peak-to-peak voltage (V_{pp}):** Defined as the difference between the maximum and minimum amplitudes of the EGM activation waveform.

(2) **Fractionation index (F):** Calculated as the number of peaks in the amplitude-normalized EGM whose absolute value exceeds a threshold of 0.3. This threshold was empirically selected to distinguish true local activations from background noise.

2.5. Statistical Analysis

For each patient, the electrophysiological markers described in the previous section were extracted. Univariate and multivariate Cox proportional hazards models were used to assess the association between these markers and AF recurrence over time, thereby accounting for the temporal nature of the outcome.

To enable comparison across variables measured on different scales, all continuous markers were standardized into clinically meaningful increments. Accordingly, hazard ratios (HR) and 95% confidence intervals (CI) are reported per unit increase of these increments. For categorical analyses, variables were dichotomized using tertile-based thresholds due to the limited cohort size and non-uniform distribution. The cutoff was defined at either the 1st or 3rd tertile, depending on the expected risk profile of each marker (e.g., the 1st tertile for CV-related variables and V_{pp}, and the 3rd tertile for all remaining variables). Two examples of Kaplan-Meier Survival Curves for CV and SCC variables are shown in Figure 1. Statistical significance was set at $p < 0.05$.

3. Results

The univariate Cox regression analysis (Table 1) identified several electrophysiological markers significantly associated with AF recurrence. Continuous CV-related variables consistently predicted recurrence, with lower values of the 10th (CV_{P10}), 25th (CV_{P25}), and 50th (CV_{med}) percentiles associated with a significant increase in recurrence risk ($p < 0.05$). Among these, CV_{med} showed the strongest categorical effect, with patients below the first

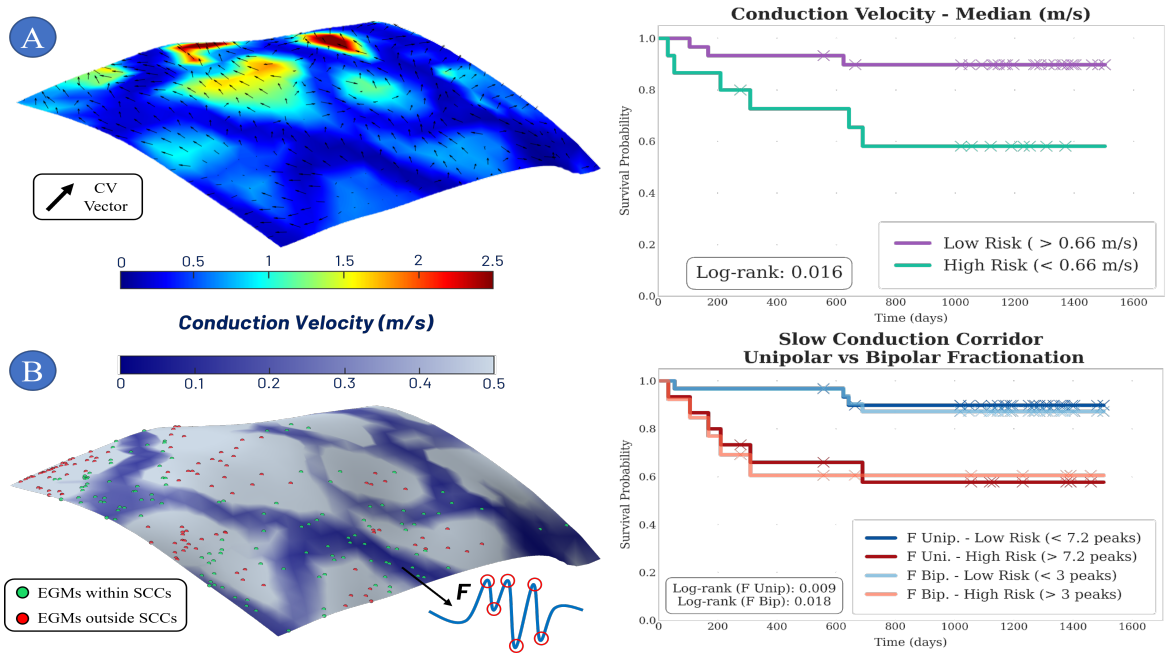


Figure 1. (A) CV and (B) SCC maps with EGM locations. Corresponding Kaplan-Meier curves show AF recurrence-free survival for markers derived from each map.

tertile threshold (0.66 m/s) showing significantly increased risk: HR of 4.69 ($p = 0.016$).

A larger SCC surface area was also associated with an increased risk of recurrence (HR: 1.81 per 5% increase, $p = 0.024$). Although total scar and border zone proportions were not significantly different in the continuous analysis, an LVZ exceeding 54.02% emerged as a strong categorical predictor (HR: 5.11, $p = 0.010$).

Signal fractionation within SCCs was also highly predictive of adverse outcomes. Fractionation in both unipolar EGMs (HR: 5.20, $p = 0.009$) and bipolar EGMs (HR: 4.39, $p = 0.018$) was among the strongest predictors of recurrence in the dichotomized analysis.

Table 2 presents the results of the multivariate Cox regression analysis. CV_{med} and SCC LVZ (%) remained significant independent predictors of AF recurrence, indicating complementary structural and functional contributions. Reduced CV_{med} below 0.66 m/s was associated with the greatest increase in risk (HR: 8.25, 95% CI: [1.78–38.24], $p = 0.007$), while an elevated LVZ area greater than 54.02% within SCCs also independently predicted recurrence (HR: 6.89, $p = 0.008$).

In an alternative model combining CV with EGM signal characteristics, both CV_{med} and SCC bipolar V_{pp} were identified as independent predictors. Specifically, reduced $CV_{med} < 0.66$ m/s and lower bipolar peak-to-peak voltage within SCCs below 0.43 mV were associated with a comparable increase in AF recurrence risk (HR: 6.37, $p = 0.011$ and HR: 6.35, $p = 0.016$, respectively).

4. Discussion

In this study, we show that CV features derived from pre-ablation electroanatomical mapping are strong predictors of long-term AF recurrence. Specifically, global conduction slowing, reflected by a lower CV_{med}, and increased signal fractionation within SCCs were both associated with a higher risk of recurrence.

Moreover, multivariate analysis revealed that combining functional (CV_{med}) and structural (low-voltage zones within SCCs) markers yields greater prognostic value than traditional mapping parameters. Notably, a CV_{med} below 0.66 m/s emerged as the most robust independent predictor of recurrence. These findings highlight the clinical relevance of CV-based substrate characterization and suggest that incorporating these metrics into clinical workflows may refine patient selection and enable more personalized ablation strategies, ultimately improving the success rates of PVI procedures.

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Table 1. Univariate Cox Analysis for Continuous and Dichotomized variables

		Continuous Analysis			Dichotomized Analysis (33rd–66th)	
	Variable	**Inc.**	**HR (95% CI)**	p**-value**	**HR (95% CI)**	**Logrank**
CV (m/s)	CV_{P10} (m/s)	-0.1	5.26 [1.15 – 25.00]	**0.031**	1.84 [0.49 – 6.86]	0.356
	CV_{P25} (m/s)	-0.1	2.77 [1.07 – 7.14]	**0.036**	2.75 [0.74 – 10.26]	0.131
	CV_{med} (m/s)	-0.1	2.08 [0.99 – 4.54]	**0.049**	4.69 [1.17 – 18.76]	**0.016**
SCC substrate and EGMs	Surface (%)	5	1.81 [1.08 – 3.03]	**0.024**	2.75 [0.74 – 10.26]	0.115
	LVZ (%)	10	1.43 [0.94 – 2.17]	0.095	5.11 [1.27 – 20.49]	**0.010**
	V_{pp} Unipolar	-0.5	2.33 [0.79 – 7.14]	0.123	3.27 [0.87 – 12.27]	0.063
	V_{pp} Bipolar	-0.5	2.50 [0.78 – 7.69]	0.124	3.24 [0.86 – 12.14]	0.066
	F Unipolar	1	1.16 [1.02 – 1.31]	**0.022**	5.20 [1.29 – 20.95]	**0.009**
	F Bipolar	1	3.03 [1.02 – 9.08]	**0.046**	4.39 [1.16 – 16.58]	**0.018**

Results are presented in Hazard Ratio (HR) and 95 % Confidence Interval (CI). CV: Conduction Velocity; SCC: Slow Conduction Corridor; LVZ: Low-Voltage Zone. Inc: Increment. Significant p-values are indicated in **bold**.

Table 2. Multivariate Cox Regression Analysis.

Model	**Variable**	**Threshold**	**HR (95% CI)**	p**-value**
Model A	CV_{med} (m/s)	< 0.66	6.37 [1.54 – 26.37]	**0.011**
	SCC LVZ (%)	> 54.02	6.89 [1.66 – 28.56]	**0.008**
Model B	CV_{med} (m/s)	< 0.66	6.37 [1.54 – 26.37]	**0.011**
	SCC bipolar V_{pp} (mV)	< 0.43	6.35 [1.41 – 28.54]	**0.016**

Results are presented in HR and 95 % Confidence Interval (CI). CV: Conduction velocity; SCC: Slow conduction corridor; LVZ: Low-voltage zone. Inc: Increment. Significant p-values are in **bold**.

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